MULTIVITAMINS

Supplement, not substitute for diet

Vitamin A in pregnancy

CHEMICAL NAME Combination preparation of fat-soluble (A, D, E, K) and water-soluble (B group, C) vitamins ± minerals (iron, zinc, iodine, selenium)

CLASS Vitamin and mineral supplement

TRADE NAMES Centrum · Pregnacare (pregnancy) · Abidec / Dalivit (children) · Cernevit / Infuvite (IV)

THERAPEUTIC USES / INDICATIONS

- Prevention of deficiency in at-risk groups: elderly, poor diet, malabsorption, alcohol misuse, chronic illness, after bariatric surgery.
- Pregnancy-specific supplements (folic acid, vitamin D, iodine).
- Infants and young children (vitamins A, C, D).
- Parenteral nutrition (IV multivitamin added to TPN).

ADMINISTRATION

- **PO** once daily with food (fat-soluble vitamins need fat).
- Children's drops/syrups measured accurately.
- IV preparations only added to infusion/TPN, not given as bolus.

DOSAGES

- **Adult:** 1 tablet daily (per product).
- **Children 6 months–5 yr:** vitamin drops (A, C, D) daily as per product.
- **Pregnancy:** pregnancy-specific product containing folic acid 400 mcg and vitamin D 10 mcg, without high-dose vitamin A.

PHARMACOKINETICS

- Water-soluble vitamins: excess excreted in urine.
- Fat-soluble vitamins: stored in liver/fat → can accumulate.

PHARMACODYNAMICS

- Replace vitamins and trace elements needed as **coenzymes and antioxidants** for metabolism, immunity, blood formation and bone health.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Mild GI upset, nausea.
- Yellow urine (riboflavin).
- Unpleasant taste.

ADVERSE EFFECTS (SERIOUS)

- Vitamin A toxicity and teratogenicity (high doses in pregnancy).
- Iron overdose in children (iron-containing products).
- Hypervitaminosis D (excess combined supplements).

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity.
- High-vitamin A products in pregnancy.
- Haemochromatosis (iron-containing products).

KEY INTERACTIONS

- **Iron/calcium/zinc content:** ↓ absorption of levothyroxine, tetracyclines, fluoroquinolones.
- **Vitamin K content:** ↓ warfarin effect (keep intake consistent).

NURSING MANAGEMENT

- Assess diet and specific deficiencies; treat specific deficits with targeted doses.
- Check products for vitamin A content in pregnancy.
- Store iron-containing products safely.

PATIENT EDUCATION

- Take with food once daily.
- Do not take more than one multivitamin product at a time.
- Pregnant women: use pregnancy-specific supplements; avoid fish liver oil and high-dose vitamin A.
- A balanced diet remains essential.

KEY POINTS

- Broad vitamin/mineral supplement.
- Avoid high vitamin A in pregnancy.
- For at-risk groups, not routine for healthy adults.
- Iron products dangerous to children.